Investigation: 80m EFHW Antenna Investigation and Troubleshooting

Callsign: AF0FR
Project: 80m End-Fed Half-Wave (EFHW) Antenna
Investigation Focus: Matching Network, Resonance Behavior, and Deployment Performance
Date: 20260507

1. Objective

The objective of this investigation was to design, build, and deploy a multiband 80m End-Fed Half-Wave (EFHW) antenna using a homebrew 49:1 transformer and evaluate its matching and resonance behavior using a NanoVNA.

Initial bench testing suggested acceptable transformer matching characteristics; however, deployed testing produced unfavorable SWR dips and resonance locations that did not align with expected amateur radio bands. This document captures the build configuration, measurements, modifications, deployment observations, and current hypotheses regarding the mismatch behavior.

2. Initial Antenna Configuration

Radiating Element

- Main element wire: 14 AWG
- Counterpoise wire: 14 AWG
- Configuration: Sloper
- Feedpoint height: ~2 meters
- Far-end height: ~20 meters

Transformer (Initial Revision)

- Transformer ratio target: 49:1
- Ferrite material: Mix 31
- Winding wire: Solid 18 AWG bifilar
- Capacitor across primary: 82 pF
- Compensation coil:
    - 6 turns
    - Located 78 inches from feedpoint

Bench Testing Method

The transformer was initially tested independently using:

- Dummy antenna resistor: 2.2 kΩ
- NanoVNA impedance measurements
- Capacitor optimization testing

Bench testing indicated the 82 pF capacitor produced the closest impedance match during isolated transformer testing.

3. Initial Transformer Investigation

Observed Issue

After deployment feedback and review of the transformer geometry, it was identified that the initial transformer winding ratio was not a true 49:1 configuration.

Actual Measured Configuration

The transformer was determined to be approximately:

- 3 primary turns
- 15 secondary turns

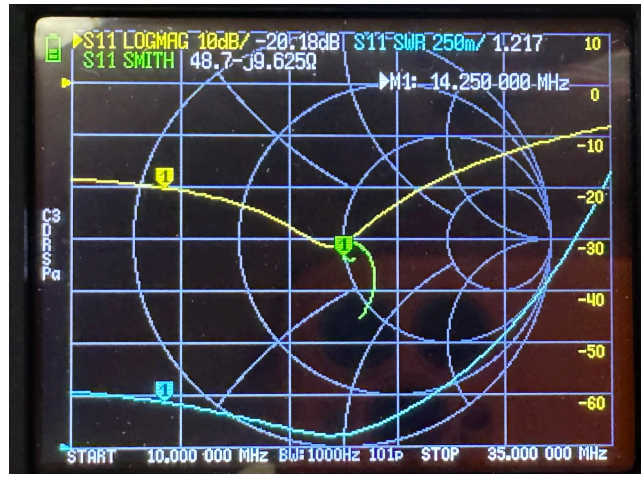
This resulted in a significantly different impedance transformation ratio than intended.

Figure #1 — Initial Transformer Construction

- 3-turn primary
- 15-turn secondary
- 82 pF capacitor across primary

The transformer was originally intended to be the correct ratio, but modified to skew measurement results.

Figure #2 — Initial Transformer Bench Measurement
Measured impedance:
Z=48.7−j9.625
This measurement appeared favorable during isolated testing with the 2.2 kΩ dummy load, leading to the assumption that the transformer design was acceptable.
However, the deployed antenna behavior later suggested this bench test did not accurately represent real antenna operating conditions.



4. Initial Deployment Results

Deployment Geometry
The antenna was deployed in a sloped configuration:

- Feedpoint approximately 2 meters above ground
- Far end approximately 20 meters above ground
- Counterpoise partially elevated during testing

The deployment was selected to allow rapid modifications and measurements.

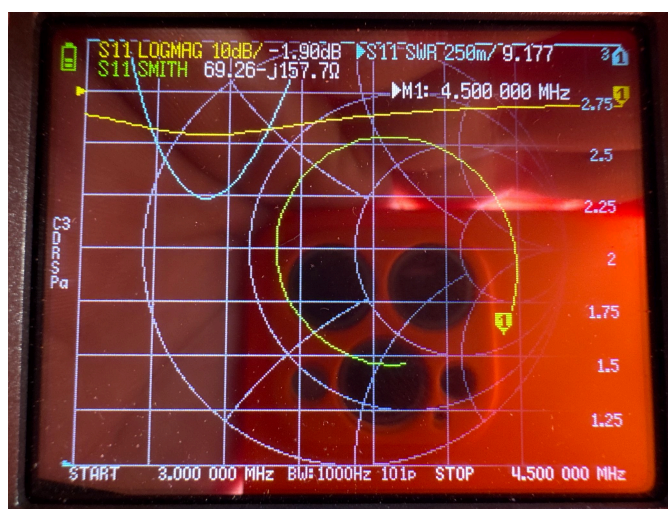
Figure #3 — Initial Broadband Sweep (80m–10m)
Initial wideband NanoVNA measurements showed:

- Multiple SWR dips
- Resonances not quite aligned with intended amateur bands
- Significant reactive behavior across much of the sweep

The 80m resonance appeared shifted upward from the intended target region.

Figure #4 — Focused 80m Measurement
Measured impedance:
Z=69.26−j157.7
Observations:

- Strong reactive component
- Poor impedance match
- Elevated SWR
- Resonance offset from expected tuning region



5. Transformer Redesign

Based on technical feedback and transformer review, the transformer was rewound to better match a true 49:1 design.

Revised Transformer Configuration

- 3 primary turns
- 21 secondary turns
- Capacitor across primary increased to 100 pF
- Dummy antenna resistor changed to 2.7 kΩ

Figure #5 — Revised 3:21 Transformer

This redesign was selected because:

- The ratio more closely approximates a true 49:1 transformation
- Additional secondary turns improve coupling on lower bands

However, isolated bench measurements appeared less favorable than the earlier transformer revision.

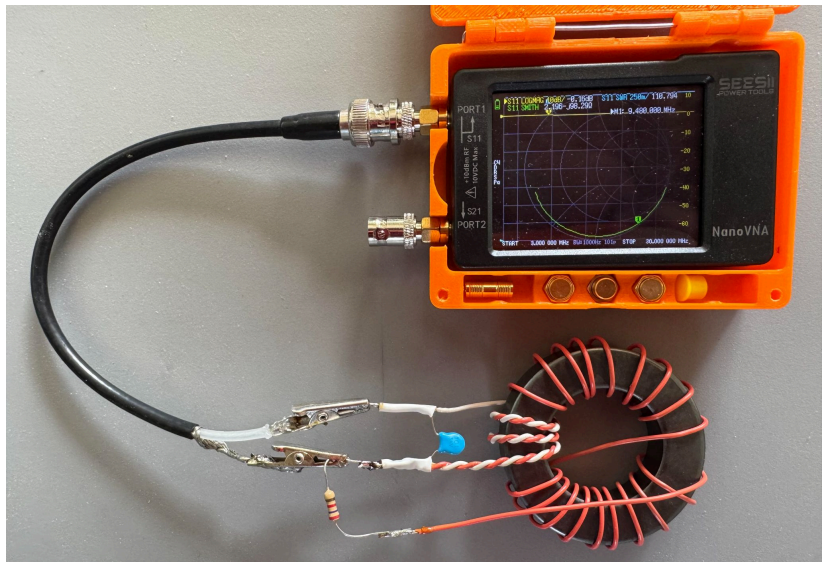
Measured impedance:

Z=2.196−j98.29

This conflicted with expectations and suggested either:

1. The isolated transformer testing method may be flawed
2. The dummy resistor does not adequately model an EFHW feedpoint
3. Additional reactive components dominate the measurement

Despite the poor isolated measurement, deployment testing continued because the transformer geometry itself was fundamentally more correct.



6. Second Deployment

Figure #6 — Revised Transformer Deployment

The revised transformer was redeployed using the same sloped antenna geometry.

No significant structural changes were made to:

- Compensation coil
- Feedpoint height
- Overall antenna length

Figure #7 — Second Deployment Measurements

Measured impedance example:

Z=8.885−j61.45

Observations:

- Large reactive component remained
- Smith chart traces remained circular and highly reactive
- SWR remained poor
- Resonance locations still did not align cleanly with expected amateur bands

The revised transformer did not significantly improve deployed performance.

7. Key Investigation Findings

7.1 Transformer Bench Testing May Be Misleading

The isolated transformer tests using fixed resistors may not accurately represent the complex impedance behavior of an actual EFHW antenna system.
The favorable 48.7 − j9.6 Ω bench result from the original transformer did not translate into favorable deployed performance.

7.2 Counterpoise Behavior Remains Uncertain

The counterpoise did not appear to significantly affect measurements when:

- partially elevated,
- partially grounded, or
- temporarily disconnected.

However, due to the end-fed nature of the antenna, counterpoise geometry may still strongly influence resonance location and feedpoint impedance.

7.3 Compensation Coil Has Not Yet Been Optimized

The compensation coil remained unchanged throughout testing.
Current configuration:

- 6 turns
- Located 78 inches from feedpoint

Potential future investigations include:

- altering turn count,
- changing spacing,
- removing the coil entirely for comparison testing.

7.4 Ferrite Material Selection May Be a Factor

The transformer currently uses Mix 31 ferrite.
Additional feedback suggested:

- Mix 31 may be better suited for common-mode chokes
- Mix 43, 52, or 61 may perform better as transformer material for a 49:1 EFHW transformer

Further transformer experimentation may be required.

8. Remaining Tuning Variables (“Knobs”)

Unlike a slim-jim antenna, where tuning parameters are visually obvious and mechanically intuitive, the EFHW contains several interacting variables that may influence resonance and matching simultaneously.
Current variables identified:

- Total radiator length
- Counterpoise length
- Counterpoise geometry
- Transformer winding ratio
- Ferrite material type
- Capacitor value across primary
- Compensation coil turn count
- Compensation coil placement
- Feedpoint height
- Antenna slope angle
- Feedline common-mode current
- Feedline interaction with antenna system

9. Current Hypotheses

The current leading hypotheses are:

1. The transformer may still not be operating correctly under real antenna conditions.
2. The compensation coil may be shifting resonances unexpectedly.
3. The deployed geometry may differ too greatly from the assumptions used during bench testing.

11. Conclusion

This configuration of the EFHW antenna system currently demonstrates unstable and poorly aligned resonance behavior despite multiple transformer revisions and capacitor adjustments.
While the revised transformer geometry is more theoretically correct, deployed measurements continue to show:

- high reactive impedance,
- poor SWR,
- and resonance locations inconsistent with intended operating bands.